

FROM UNDERSTANDING TO ACCURACY

FOUR PROBLEM AREAS

✗ Direction Error

✗ Logic Error

✗ Operator Mixing

✗ Terminology Error

THE IMPROVEMENT LEARNING CYCLE

1. UNDERSTAND

2. WRITE

3. PRACTICE

4. MAKE MISTAKES

5. IDENTIFY WHY

6. CORRECT

7. PRACTICE AGAIN

8. RECALL

9. AUTOMATIC

YOUR FOUR IMPROVEMENT AREAS

IMPROVEMENT AREAS

1. Comparison Direction

> < >= <=

Problem: left/right direction sometimes gets reversed.

2. Logical Operators

&& || !

Mental shortcuts: && = ALL, || = ANY, ! = NOT

3. Operator Categories

! != !== =

Can sometimes get mixed.

4. Exact Technical Terminology

Concept is understood, but technical definitions can sometimes be inaccurate.

COMPARISON DIRECTION

>

<

>=

<=

CORE PROBLEM: Left / Right Direction

 $A > B$

Is A greater than B?

 $8 > 5$

Is 8 greater than 5?

True

COMPARISON OPERATORS: CORE RULE

A > B



A is greater than B.

A < B



A is less than B.

A >= B



A is greater than or equal to B.

A <= B



A is less than or equal to B.

COMPARISON DIRECTION NOTES

CONCEPT: Comparison Operators

RULE: Always read the expression as a sentence.

EXAMPLE: $8 > 5$

QUESTION: Is 8 greater than 5?

ANSWER: True

More Examples

$3 > 7 \rightarrow$ Is 3 greater than 7? $\rightarrow$ False

$4 \leq 4 \rightarrow$ Is 4 less than or equal to 4? $\rightarrow$ True

Source: Section 3

COMPARISON DIRECTION DRILL

LEVEL 1 — Recognition

$>$ $<$ $>=$ $<=$

LEVEL 2 — Sentence Conversion

$9 > 4 \rightarrow$ Is 9 greater than 4?

LEVEL 3 — Truth Evaluation

$9 > 4 \rightarrow$ Is 9 greater than 4? $\rightarrow$ True

MASTERY RULE: $A > B \rightarrow$ Is A greater than B?

Until this can be done without thinking, speed does not need to increase.

LOGICAL OPERATORS

LOGICAL OPERATORS

&&

||

!

ALL

ANY

NOT

&& LOGICAL AND

&&

Logical AND

The logical AND operator evaluates to true only when both operands are true.

Mental Model: ALL

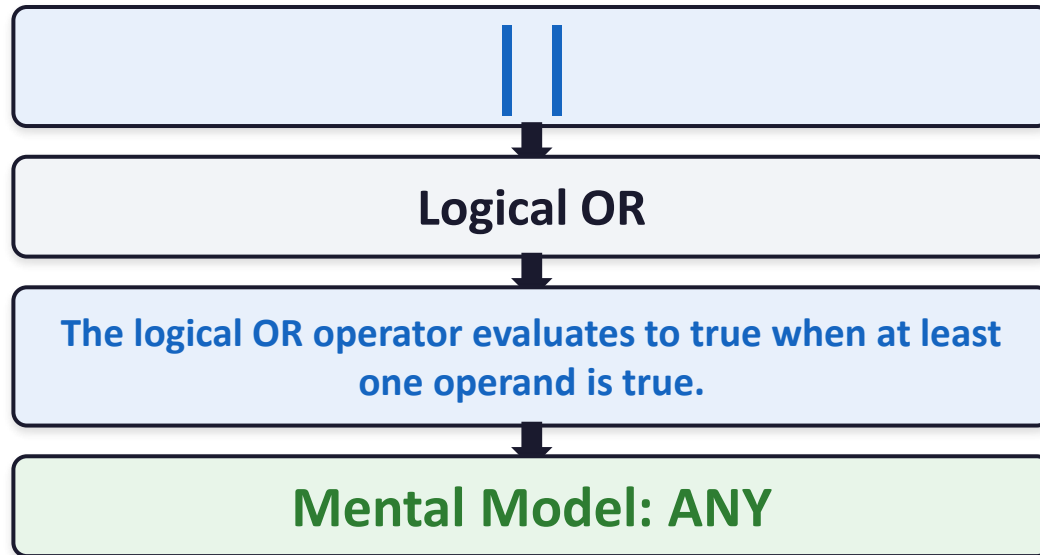
true && true

→ Both are true. → **true**

true && false

→ All true नहीं हैं। → **false**

|| LOGICAL OR



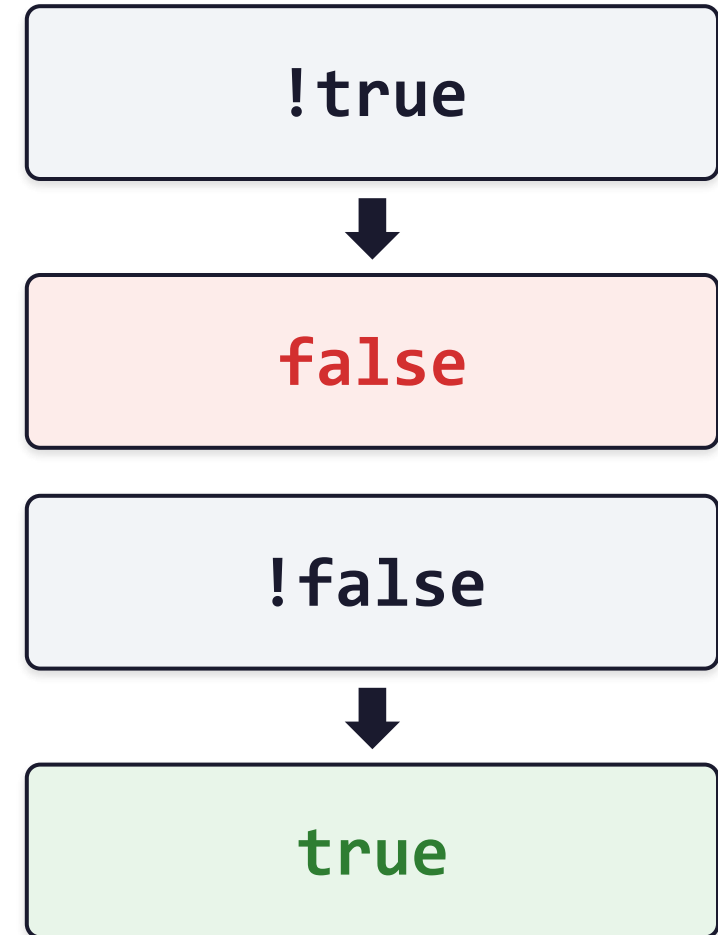
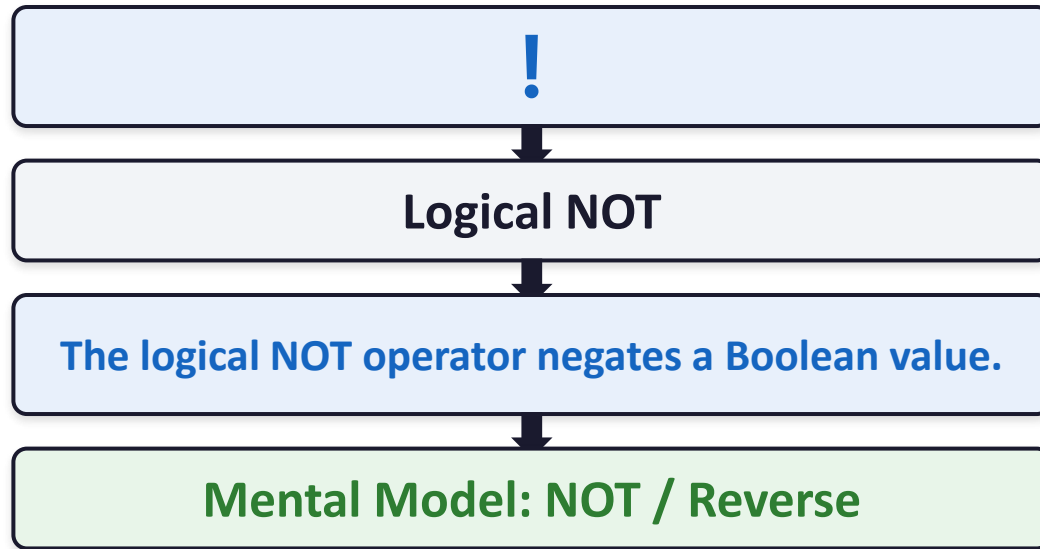
`true || false`

→ At least one true है। → **true**

`false || false`

→ कोई भी true नहीं है। → **false**

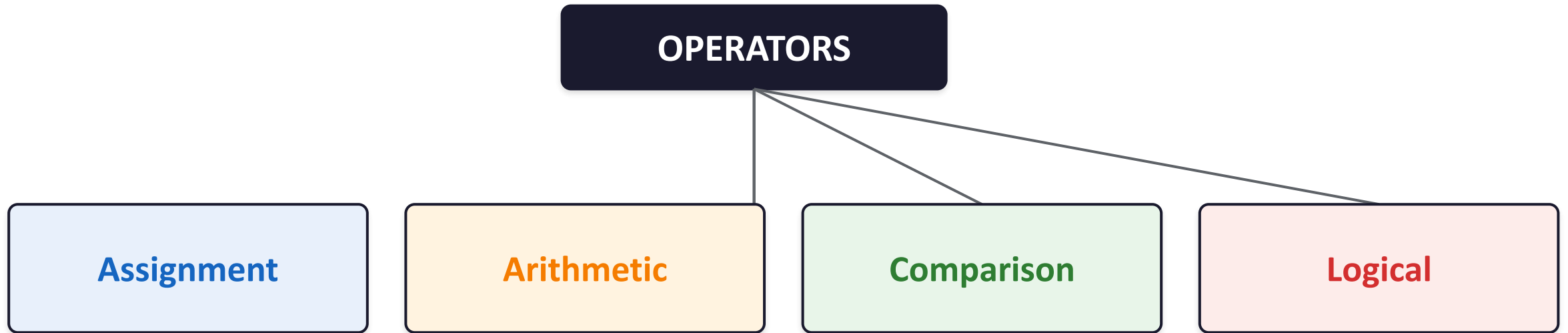
! LOGICAL NOT



LOGICAL OPERATORS — WORKING NOTES

CONCEPT	Logical AND (&&)
Technical Definition	Returns true only when both operands are true.
Mental Shortcut	ALL
Example	<code>true && false</code>
Result	false
My Explanation	Both conditions must be true.

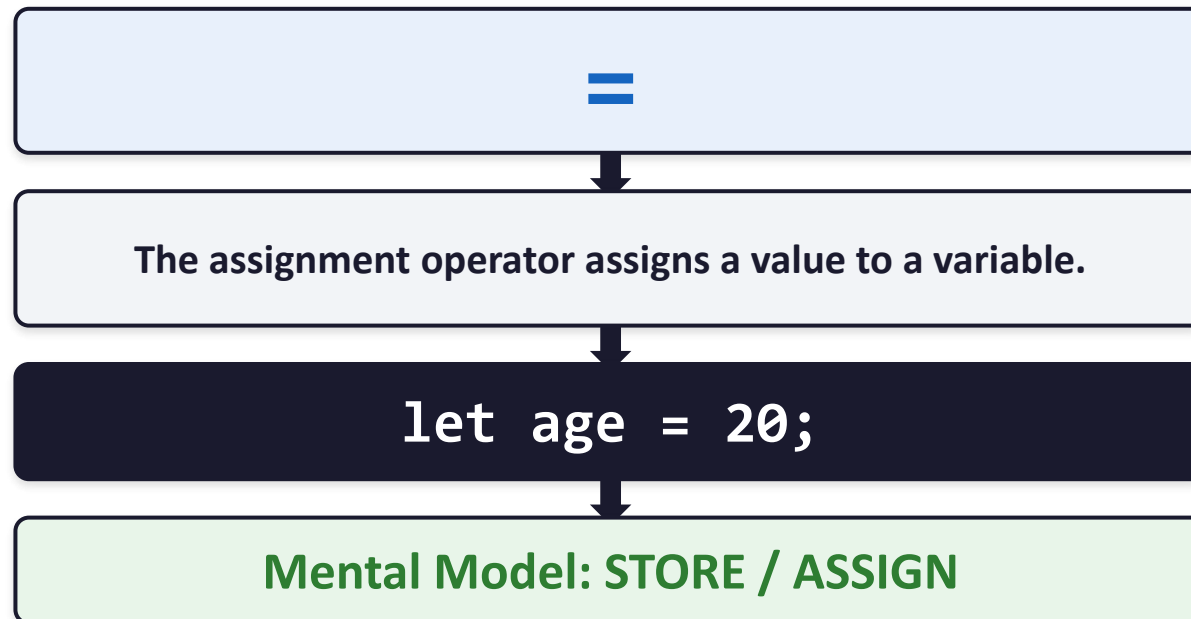
OPERATOR CATEGORIES



FOUR OPERATOR FAMILIES

Family	Operators	Mental Model
Assignment	=	STORE / ASSIGN
Arithmetic	+ - * / %	CALCULATE
Comparison	> < >= <= == != === !==	COMPARE
Logical	&& !	COMBINE / NEGATE CONDITIONS

ASSIGNMENT OPERATOR



ARITHMETIC OPERATORS

+ **-** **×** **÷** **%**



CALCULATE

5 + 2

5 - 2

5 * 2

5 / 2

5 % 2

COMPARISON OPERATORS

> < >= <= == != === !==



COMPARE

5 > 3

5 != 3

5 === 5

5 !== "5"

LOGICAL OPERATORS

&&

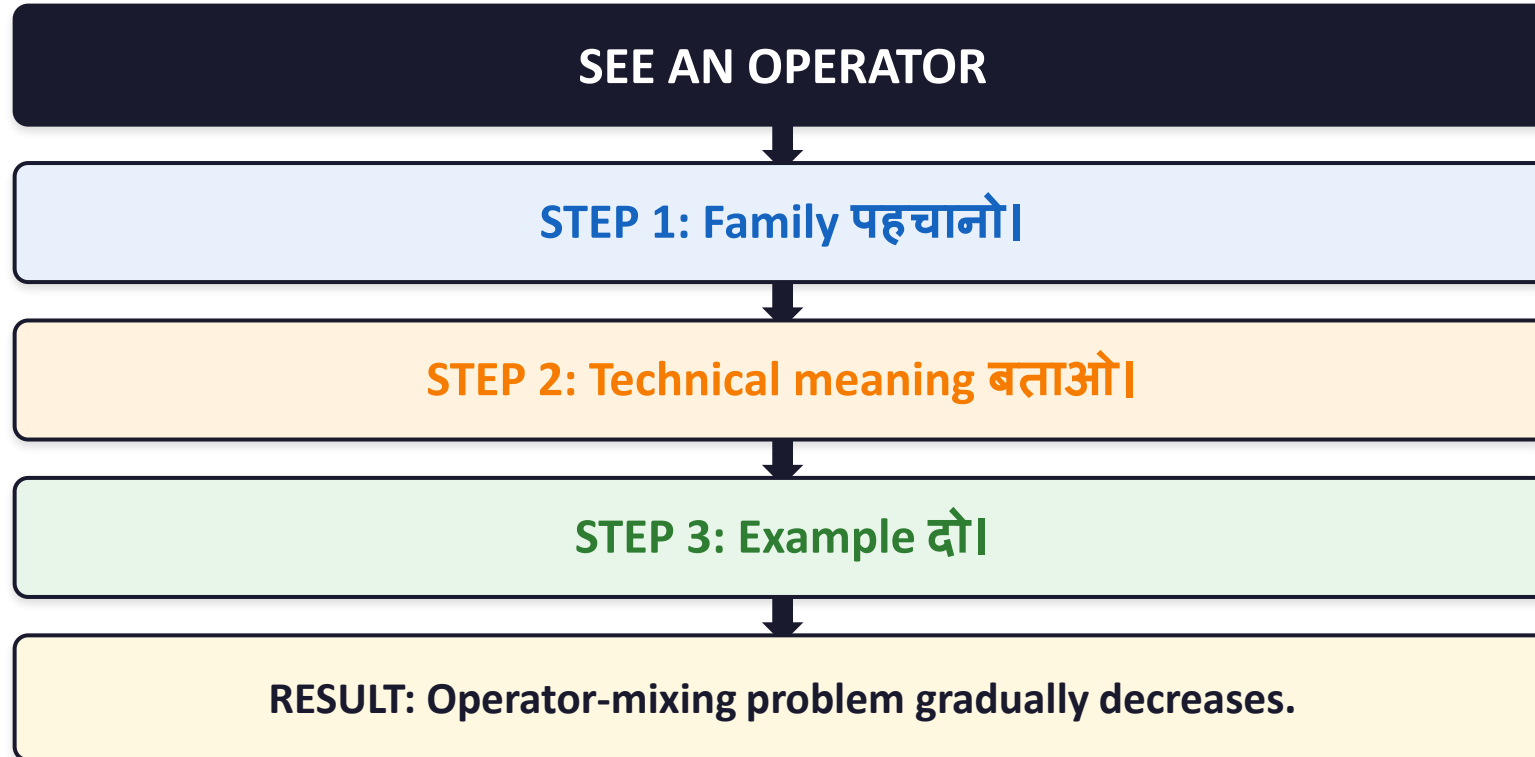
||

!



COMBINE / NEGATE CONDITIONS

OPERATOR FAMILY NOTEBOOK



EXACT TECHNICAL TERMINOLOGY

Understanding a concept $\neq$ accurately stating its technical definition

1. Technical Definition

2. Example

3. My Explanation

LAYER 1: Technical Definition



LAYER 2: Simple Explanation



LAYER 3: Example

ASSIGNMENT OPERATOR

DEFINE → SHOW → EXPLAIN

DEFINE

The assignment operator assigns a value to a variable.

SHOW

```
let x = 10;
```

EXPLAIN

x के अंदर 10 store करना।

LOGICAL AND

TECHNICAL → EXAMPLE → PERSONAL EXPLANATION

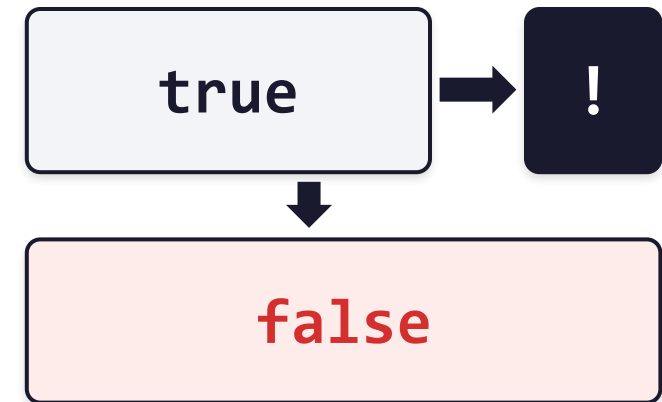
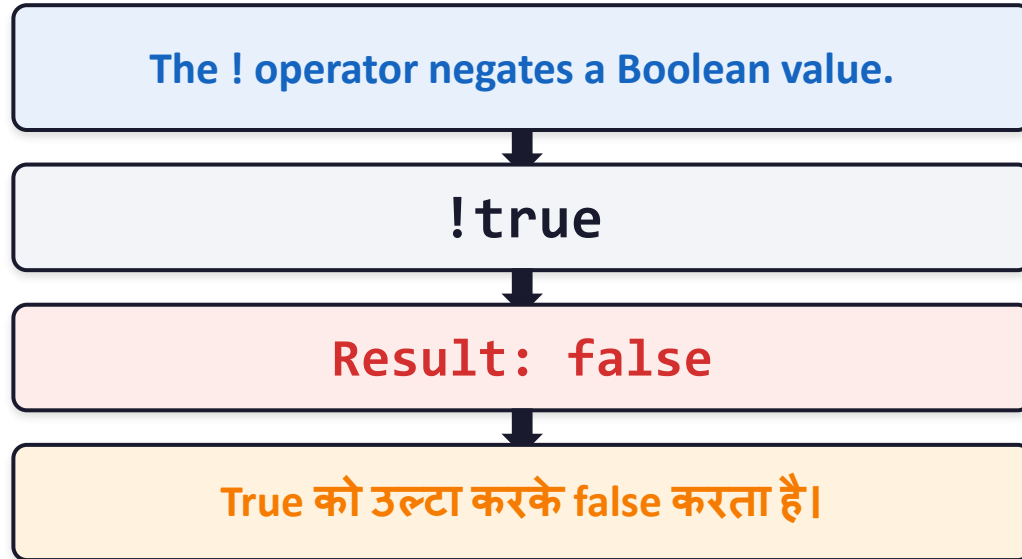
The && operator returns true only when both operands are truthy.

`true && true`

Result: true

दोनों conditions pass होनी चाहिए।

LOGICAL NOT



WORKING NOTES TEMPLATE

1. Technical Definition

2. Simple Meaning

3. Mental Shortcut

4. Syntax / Operator

5. Example

6. Result

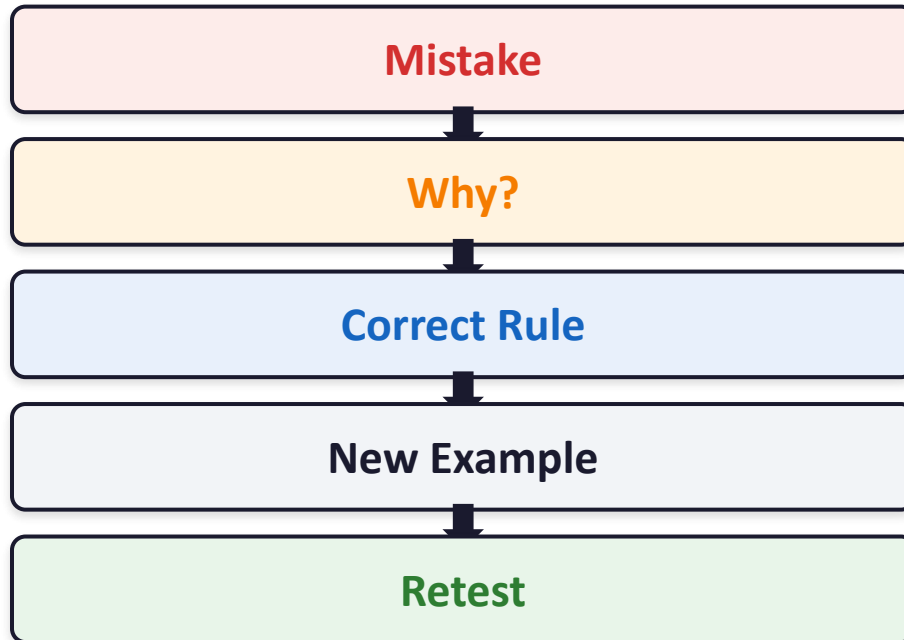
7. Why?

8. Common Mistake

9. My Explanation

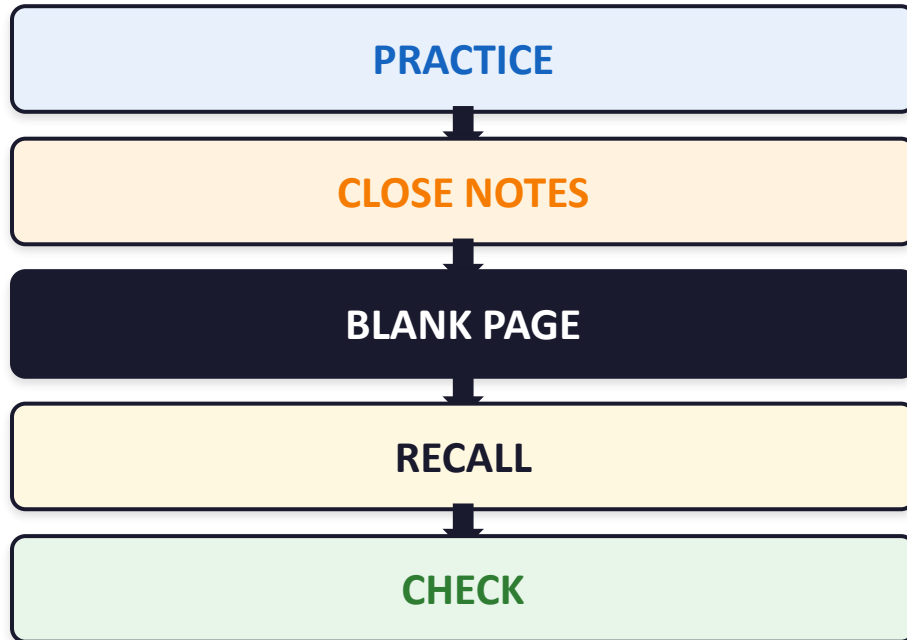
10. One Recall Question

ERROR LOG: THE MOST IMPORTANT PART



QUESTION:	5 > 8
MY ANSWER:	True
CORRECT ANSWER:	False
WHY WAS I WRONG?	I reversed the comparison direction.
CORRECT CONCEPT:	Ask: "Is 5 greater than 8?"
PREVENTION:	Convert every comparison into a sentence.

ACTIVE RECALL METHOD



Recall Questions

1. What does $>$ mean?
2. What does $<$ mean?
3. What does $\geq$ mean?
4. What does $\leq$ mean?
5. How do I avoid direction mistakes?

PRACTICE LEVELS

LEVEL 1: Recognition

`&&` → Logical operator

LEVEL 2: Meaning

`&&` → AND / ALL

LEVEL 3: Application

`true && false` → `false`

LEVEL 4: Explanation

Why is the answer false?

LEVEL 5: Technical Definition

Define the `&&` operator technically.

MIXED PRACTICE

Q1. What family does = belong to?

Q2. What does && mean?

Q3. Is $7 > 10$ true or false?

Q4. What does ! do?

Q5. What is the difference between != and !==?

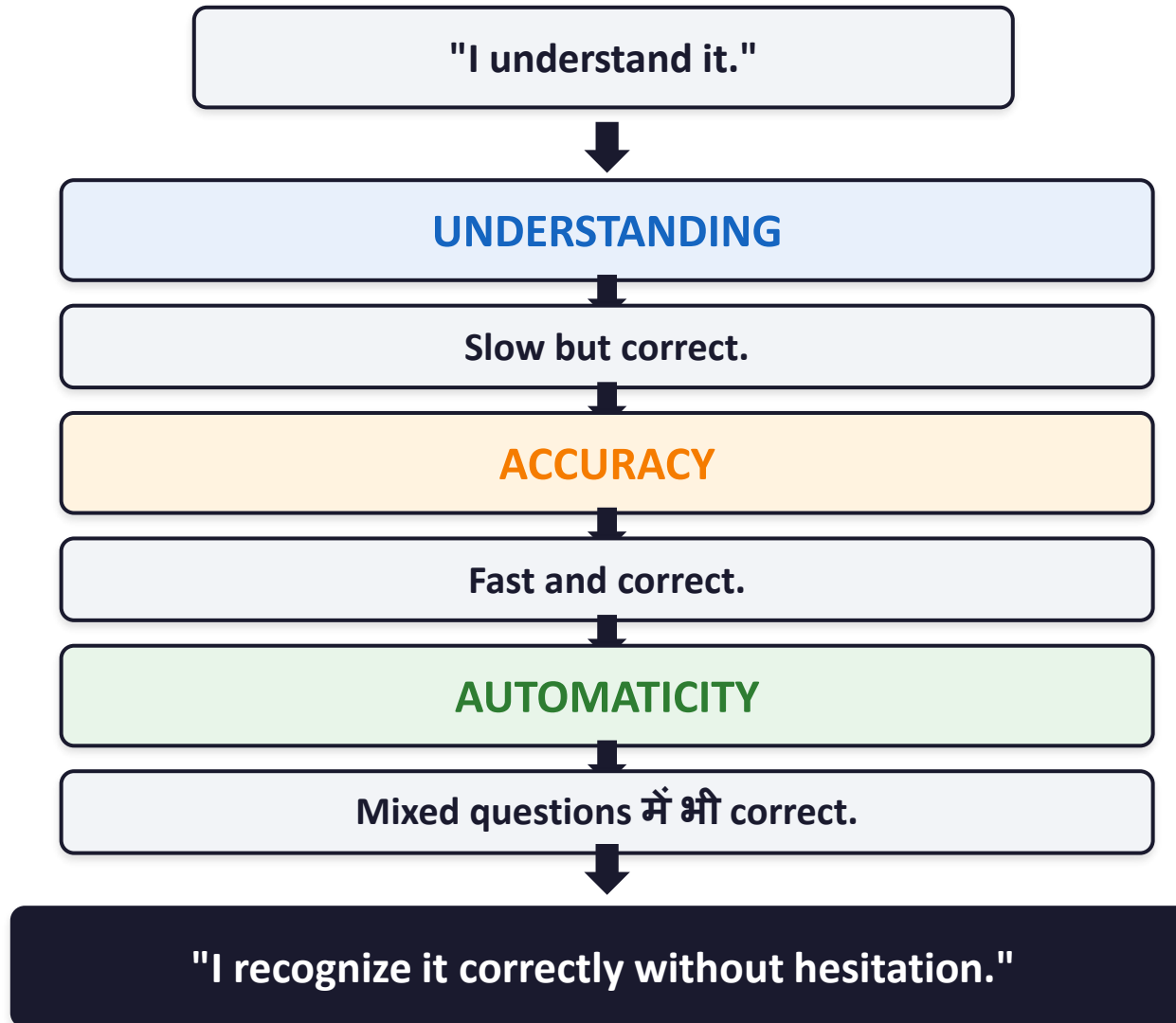
Q6. Give a technical definition of ===.

Q7. What family does !== belong to?

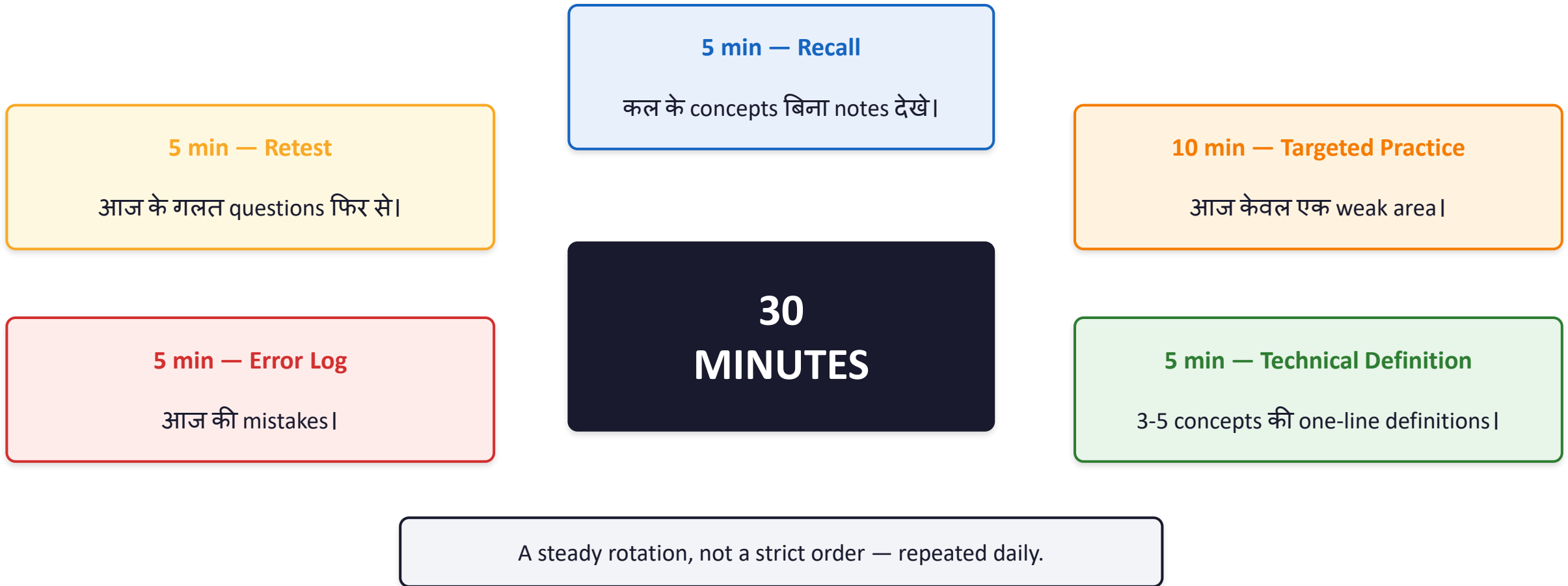
Q8. Evaluate: true || false

CONTEXT SWITCHING

AUTOMATICITY TRAINING



DAILY 30-MINUTE IMPROVEMENT SYSTEM



WEEKLY IMPROVEMENT CYCLE

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
Comparison	Logical Operators	Operator Categories	Technical Terminology	Mixed Practice	Error Retest	Mastery Test

If one area has more errors, increase its practice volume next week.

MASTERY TRACKING

Area	Questions	Score	Status
Comparison	50	94%	 Strong/Mastered
Logical	50	86%	 Improving
Categories	50	90%	 Strong/Mastered
Terminology	40	75%	 Weak
 < 80% Weak	 80–89% Improving	 90–94% Strong	 95%+ Mastered

MASTERY TEST

COMPARISON

- ✓ direction सही पढ़ो
- ✓ sentence correctly बनाओ
- ✓ True/False सही निकालो

LOGICAL

&& || !

CATEGORIES

- ✓ किसी operator की family तुरंत पहचानो।

TERMINOLOGY

- ✓ 1 technically accurate definition + 1 correct example

ONE-PAGE IMPROVEMENT MASTER SHEET

1. COMPARISON

> greater than
< less than
>= greater/equal
<= less/equal
Rule: "Is A ... B?"

2. LOGICAL

&& = ALL
|| = ANY
! = NOT

3. OPERATOR FAMILIES

= Assignment
+ - * / % Arithmetic
> < >= <= Comparison
&& || ! Logical

4. TECHNICAL DEFINITION

CONCEPT → Definition → Example
→ My Explanation

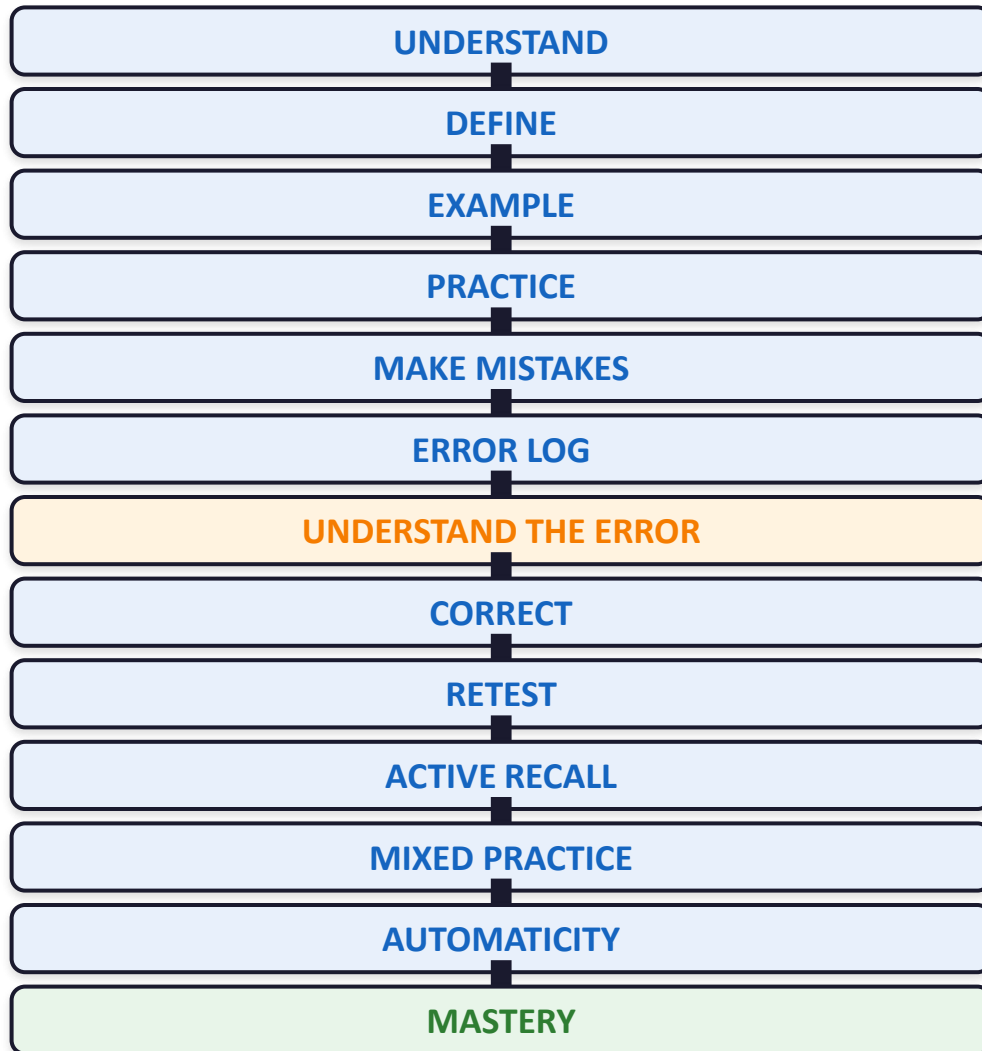
5. ERROR FORMULA

Mistake → Why? → Correct Rule
→ New Example → Retest

6. MASTERY

Understand → Practice → Recall
→ Correct → Repeat → Automatic

THE GOLDEN RULE



गलती होना problem नहीं है। एक ही गलती को बिना समझे दोबारा करना problem है।

Error Log = Personal Improvement Database